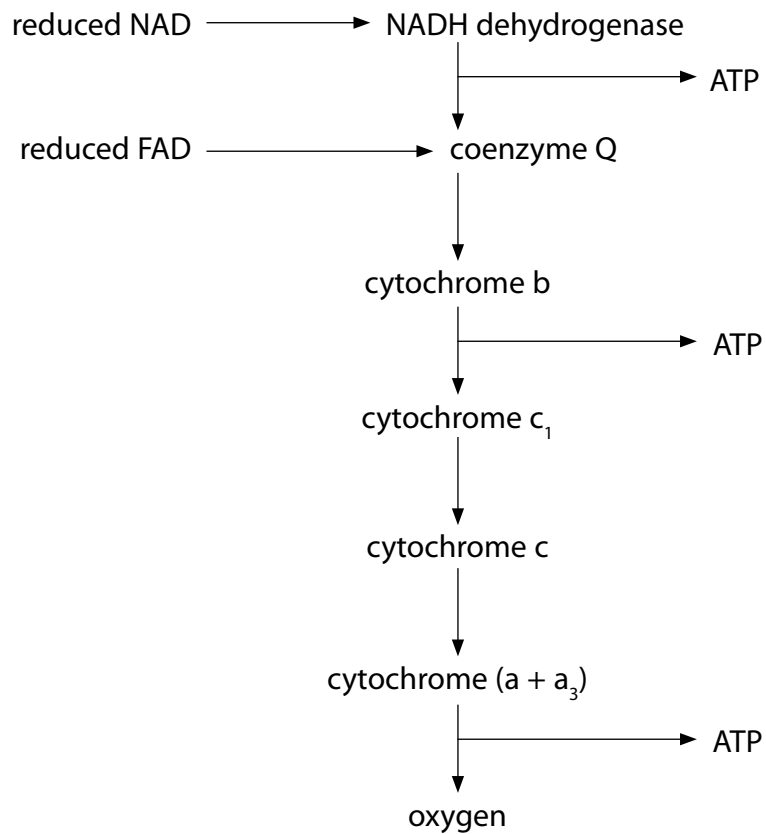


- 1 The diagram shows the sequence of electron carriers in the electron transport chain. The diagram also shows the stages where ATP is generated.



(a) Where is the site of the electron transport chain?

(1)

- ☐ A cytoplasm
- ☐ B inner mitochondrial membrane
- ☐ C matrix
- ☐ D outer mitochondrial membrane

(b) The table gives some information about inhibitors of the electron transport chain.

Inhibitor	Site of inhibition
Cyanide	between cytochrome (a + a ₃) and oxygen
Rotenone	between NADH dehydrogenase and coenzyme Q

- (i) Explain the effect that cyanide will have on ATP production by the electron transport chain.

(2)

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- (ii) Metabolism of succinate results in the production of reduced FAD.

Explain the effect that rotenone will have on ATP production by the electron transport chain.

(2)

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(iii) Explain why these inhibitors will not affect the production of ATP from anaerobic respiration. Use the information in the table to support your answer.

(2)

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(Total for Question 1 = 7 marks)